

2nd Set of Problems

1. Solve the direct kinematics for the manipulator illustrated in Figure 1. In other words, determine the last frame's position and orientation with respect to frame 0 (See the Appendix for the joint angle convention).

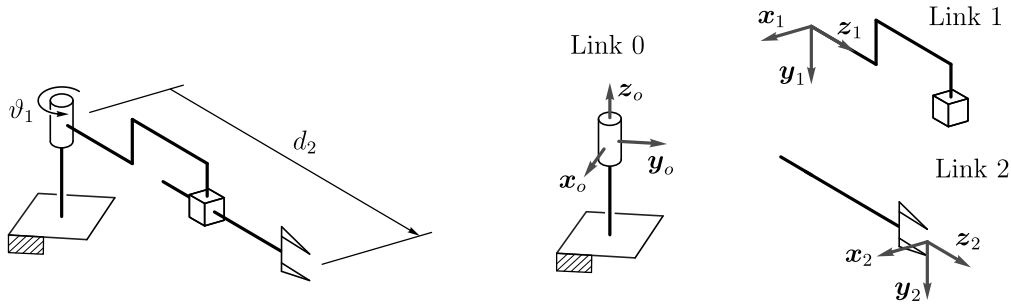


Figure 1: Polar manipulator. Its reference frames have been placed, at each link, according to the Denavit-Hartenberg convention.

2. Solve the direct kinematics for the manipulator illustrated in Figure 2.

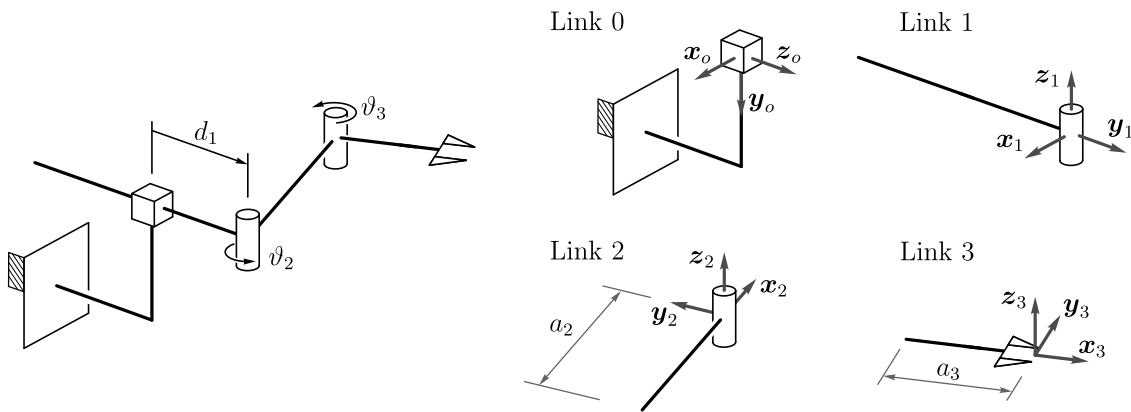


Figure 2: Stadium manipulator. Its reference frames have been placed according to the Denavit-Hartenberg convention.

3. Solve the direct kinematics for the manipulator illustrated in Figure 3(a).
4. Solve the direct kinematics for the manipulator illustrated in Figure 3(b).
5. For a given application, suppose that the orientation of the manipulator's end-effector is known with respect to the base, through

$$\mathbf{R}_e^b = \begin{pmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{pmatrix} = \begin{pmatrix} 0.9659 & 0.2241 & 0.1294 \\ 0.1830 & -0.2380 & -0.9539 \\ -0.1830 & 0.9451 & -0.2709 \end{pmatrix}.$$

Suppose also that a XYX sequence of rotations is desired to express the manipulator's rotation.

- Illustrate the sequence of rotations to write R_e^b as a function of the sequence's Euler angles, φ , θ and ψ .
- Solve the inverse problem to verify that $\varphi \approx \pi/4$, $\theta \approx \pi/12$ and $\psi \approx \pi/3$, or $\varphi \approx \pi/4 - \pi$, $\theta \approx -\pi/12$ and $\psi \approx \pi/3 - \pi$.

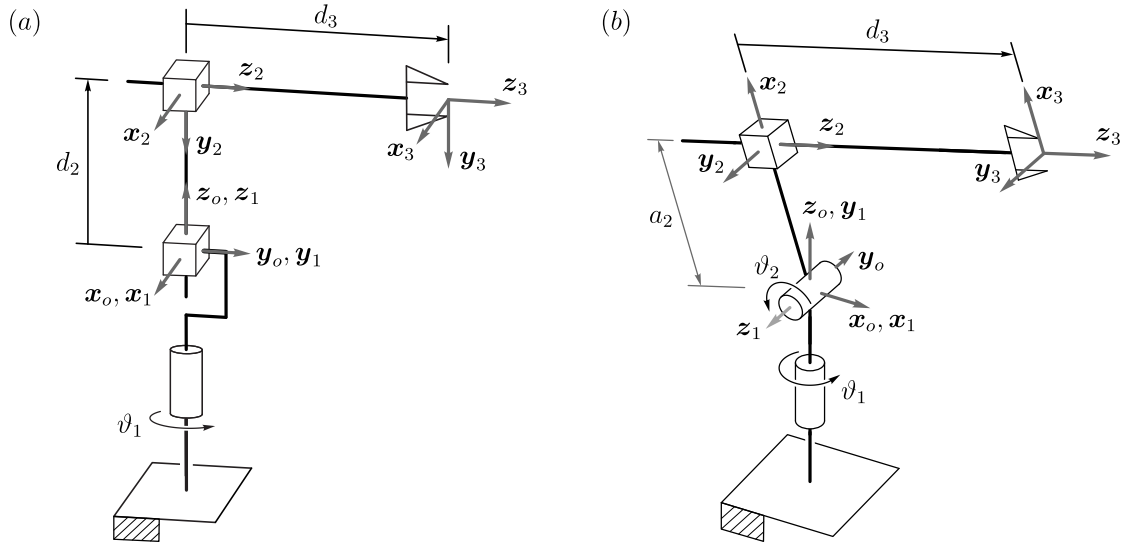


Figure 3: (a) Cylindrical and (b) Spherical manipulators. Their reference frames have been placed according to the Denavit-Hartenberg convention.

Appendix

Given a joint rotation between two frames, where i is the antecedent and j the precedent frame, its direction is measured positive or negative with respect to axis z_i , according to the right-hand rule, and its amplitude is measured between axes x_i and x_j . Said joint angle is notated as ϑ_j and drawn with an arrowhead with endpoint in x_j , as illustrated in Figure 4.

Angle amplitudes, such as α , β , η and γ in Fig. 4, are notated without arrowhead and always assumed positive.

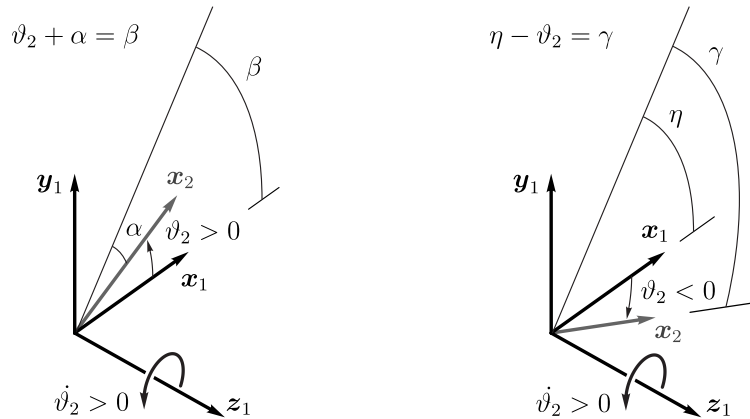


Figure 4: Angle nomenclature.